Tooth Number Detection with YOLO11

Short project report

Objective

Train a YOLO-based detector that identifies and localizes 32 tooth-number classes in panoramic dental images.

Dataset

- Images used: 497
- Bounding boxes: 13848
- Split: train 347, val 74, test 76

Approach

- Mixed YOLO box and polygon annotations were normalized into detection boxes.
- Ultralytics YOLO11m was fine-tuned with transfer learning.
- Input size was increased to 1024 to improve tight localization on dense tooth structures.
- Flips and anatomy-breaking augmentations were disabled because tooth numbering depends on left-right consistency.

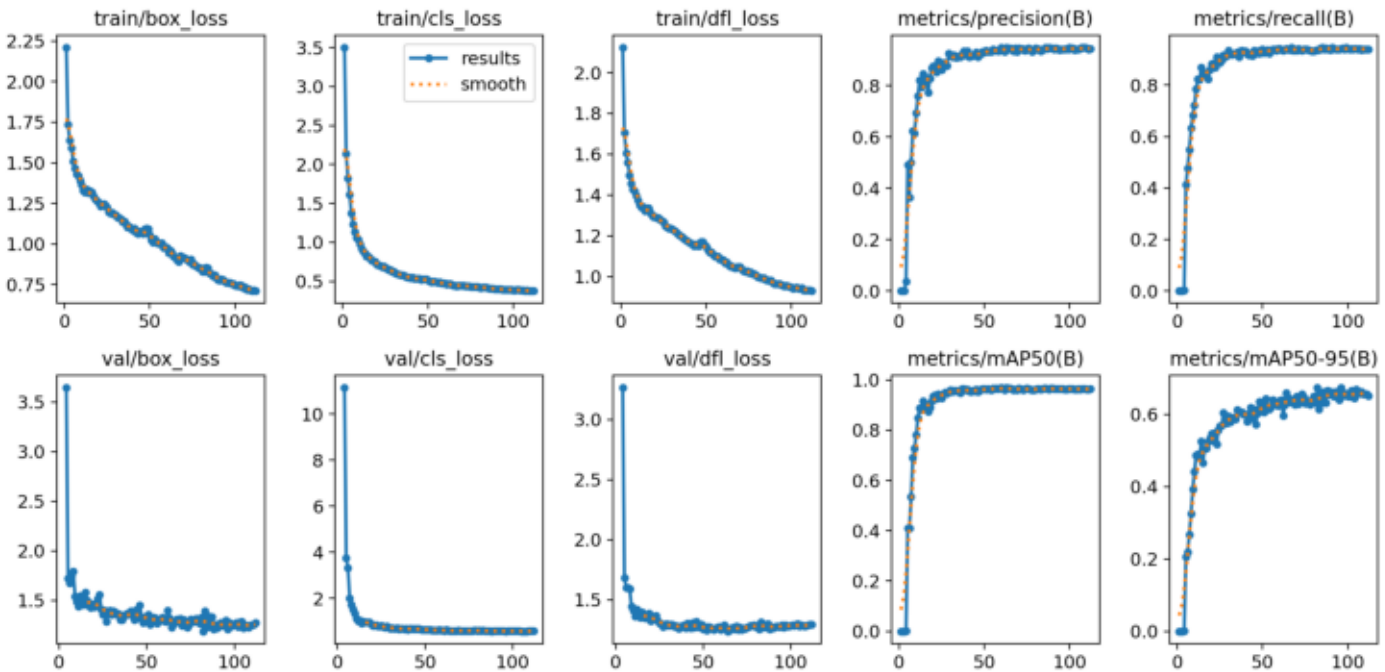
Training setup

- Model: yolo11m.pt
- Epochs: 120
- Image size: 1024
- Batch size: 6
- Optimizer: AdamW

Final metrics

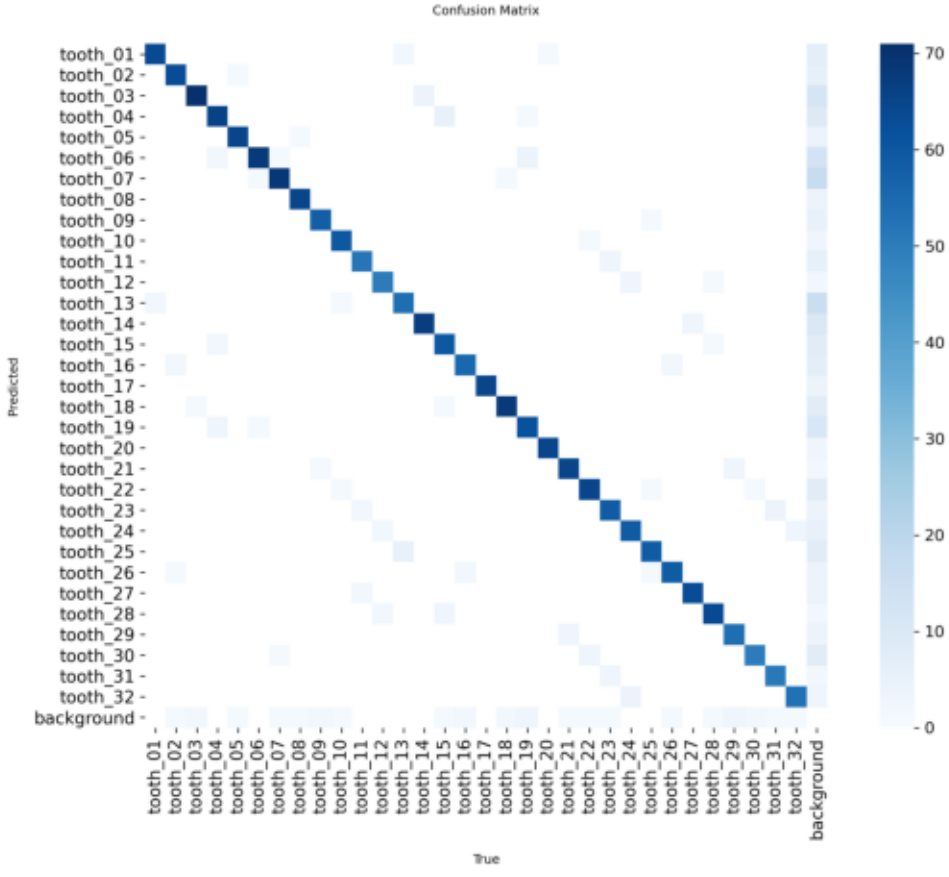
- Validation precision / recall: 0.9412 / 0.9357
- Validation mAP50 / mAP50-95: 0.9638 / 0.6745
- Test precision / recall: 0.9389 / 0.9498
- Test mAP50 / mAP50-95: 0.9698 / 0.6589

Training curves



Confusion matrix and sample inference

Confusion matrix



Sample prediction output

Input image and sample prediction

Input panoramic image

Predicted tooth numbering output